Calculation (linearisation)

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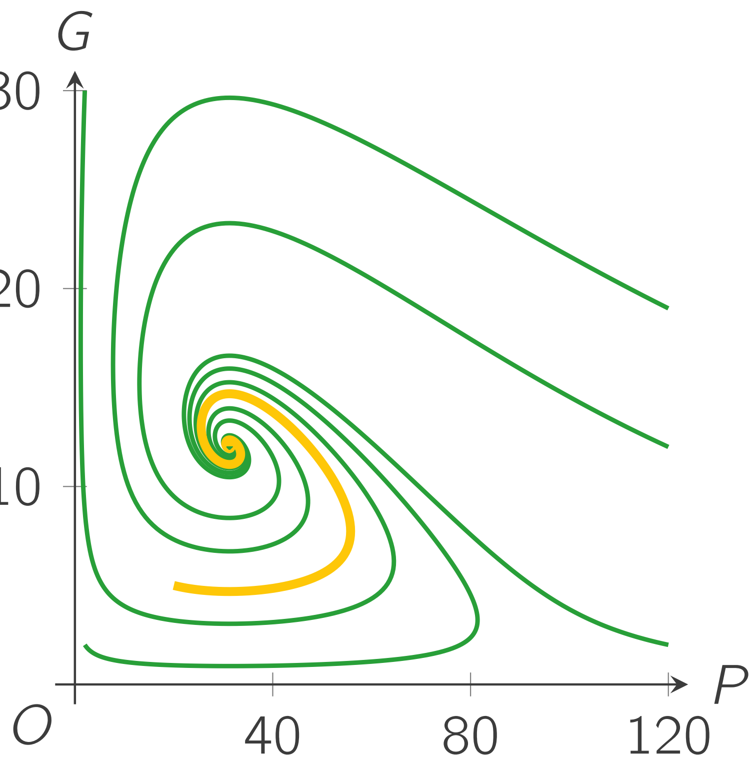
A mathematical model for interacting populations of rainbowfish $(P(t))$ and gourami $(G(t))$ in a 100-rainbowfish aquarium is the following system of differential equations.

$$\begin{cases} \dfrac{dP}{dt} &= 0.7\,P(t) - 0.007\,P^2(t) - 0.04\,P(t)G(t), \\[2ex] \dfrac{dG}{dt} &= -0.25\,G(t) + 0.008\,P(t)G(t). \end{cases}$$

You have learned to simulate solutions with Euler's method and to visualise these in the phase plane. Here you see a combination of trajectories, each with a different set of initial values.



Can we calculate some results analytically, so we can understand better how the parameters we have chosen for the model influence the results?

In this section you will learn how you to linearise the system in the neighbourhood of an equilibrium point. With the linearisations, you can predict some of the results analytically from the parameters.

The definition of an equilibrium point is:

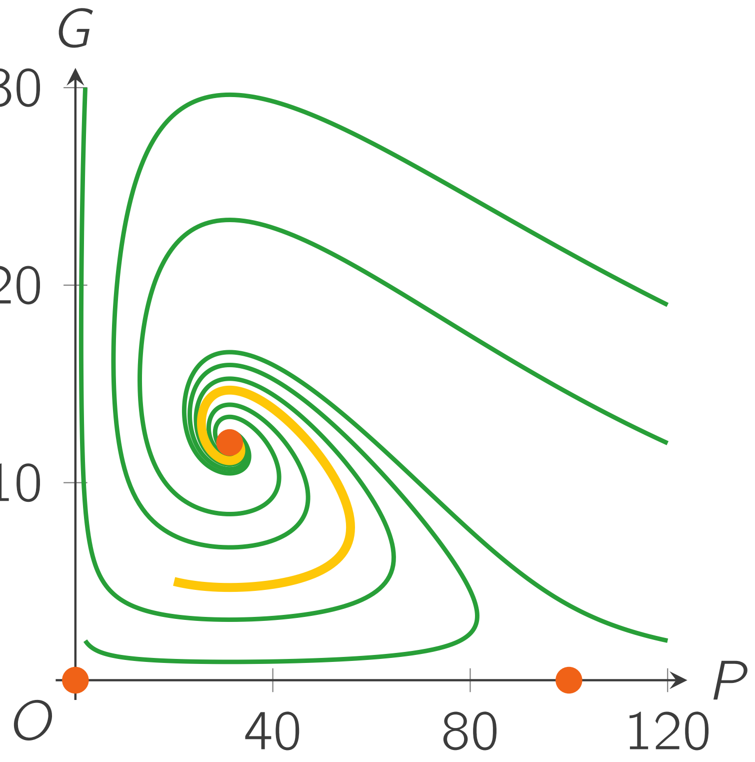
An equilibrium point of a system of differential equations

$$\frac{d\vec{X}}{dt} = \vec{F}(\vec{X}),$$

is a point $\vec{X}_0$ *where*

$$\frac{d\vec{X}_0}{dt} = \vec{F}(\vec{X}_0) = \vec{0}.$$

In §3.1 Calculation (phase plane), you have calculated the equilibrium points for this system: $(0,0)$, $(100,0)$ and $(31.25, 12.03125)$. Adding these equilibrium points to the phase portrait gives:



You can see special things happening near the equilibrium points. For example near the point $(100,0)$, you can see lines going first towards and then away from the equilibrium. Could we have predicted this behaviour using just the differential equations?

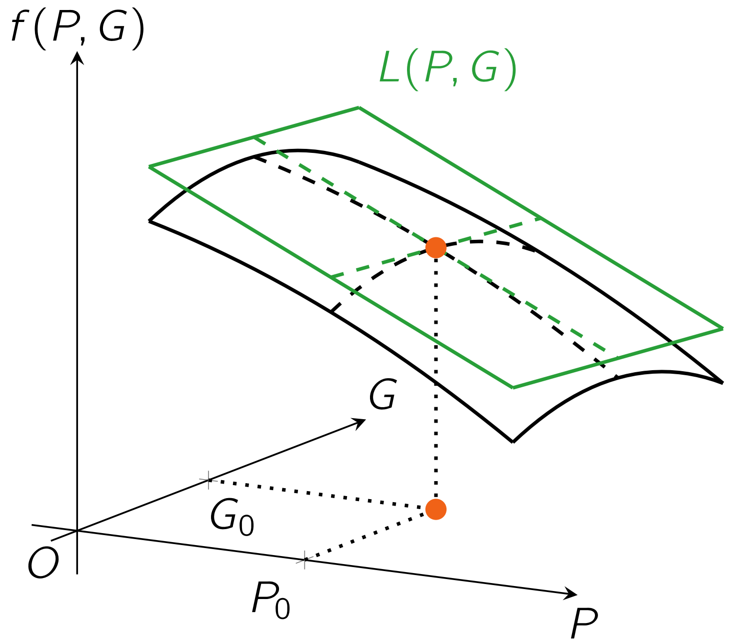
The answer is "Yes", and in the remainder of this section you will learn how you can do this.

The first step in obtaining a prediction of the behavior near the equilibrium point is the *linearisation* of the differential equations around the equilibrium point.

The linearisation of a function $f(P, G)$ of two variables P and G is the process of approximating this function f in the neighbourhood of the point (P_0, G_0) by a function $L(P, G)$ with the formula:

$$L(P,G) = f(P_0,G_0) + \left.\frac{\partial f}{\partial P}\right|_{(P_0,G_0)} (P - P_0) + \left.\frac{\partial f}{\partial G}\right|_{(P_0,G_0)} (G - G_0).$$

$L(P, G)$ is the plane tangent to f in the point $(P_0, G_0, f(P_0, G_0))$. A visualisation of such a tangent plane is:



In our case we have two differential equations, and two functions to linearise. In a general form:

$$\begin{cases} \dfrac{dP}{dt} & = & f_1(P,G), \\ \dfrac{dG}{dt} & = & f_2(P,G). \end{cases}$$

We can linearise the two right-hand-side functions f_1 and f_2 around the same equilibrium point (P_0, G_0), which leads to the general system

$$\begin{cases} \dfrac{dP}{dt} & = & \left.\dfrac{\partial f_1}{\partial P}\right|_{(P_0,G_0)} (P-P_0) + \left.\dfrac{\partial f_1}{\partial G}\right|_{(P_0,G_0)} (G-G_0), \\ \dfrac{dG}{dt} & = & \left.\dfrac{\partial f_2}{\partial P}\right|_{(P_0,G_0)} (P-P_0) + \left.\dfrac{\partial f_2}{\partial G}\right|_{(P_0,G_0)} (G-G_0). \end{cases}$$

Note that the terms $f_1(P_0, G_0)$ and $f_2(P_0, G_0)$ have been left out, because both are zero in an equilibrium point by definition.

Exercise A

1/1 point (graded)

Linearise our system of differential equations around $(P_0, G_0) = (100, 0)$.

$$\begin{cases} \dfrac{dP}{dt} = 0.7\,P - 0.007\,P^2 - 0.04\,PG, \\ \dfrac{dG}{dt} = -0.25\,G + 0.008\,PG. \end{cases}$$

What is the correct system after linearisation around $(100, 0)$?

- ( ) $$\begin{cases} \dfrac{dP}{dt} = (0.7 - 0.014\,P - 0.04\,G)\,(P-100) - 0.04\,P\,G, \\ \dfrac{dG}{dt} = 0.008\,G\,(P-100) + (0.008\,P - 0.25)\,G. \end{cases}$$

- ( ) $$\begin{cases} \dfrac{dP}{dt} = -0.7\,(P-100), \\ \dfrac{dG}{dt} = -4\,(P-100) + 0.55\,G. \end{cases}$$

- ( ) $$\begin{cases} \dfrac{dP}{dt} = (0.7 - 0.014\,P - 0.04\,G)\,(P-100) + 0.008\,G^2, \\ \dfrac{dG}{dt} = -0.04\,P\,(P-100) + (0.008\,P - 0.25)\,G. \end{cases}$$

- (•) $$\begin{cases} \dfrac{dP}{dt} = -0.7\,(P-100) - 4\,G, \\ \dfrac{dG}{dt} = 0.55\,G. \end{cases}$$ ✔

Submit You have used 2 of 2 attempts

Answers are displayed within the problem

We can rewrite this system of linearised differential equations into a different form by using vectors.

The first step is to write, just like how we did for Euler's method, the linearised system in vectors:

$$\begin{bmatrix} \dfrac{dP}{dt} \\ \dfrac{dG}{dt} \end{bmatrix} = \begin{bmatrix} \left.\dfrac{\partial f_1}{\partial P}\right|_{(P_0,G_0)} (P - P_0) + \left.\dfrac{\partial f_1}{\partial G}\right|_{(P_0,G_0)} (G - G_0) \\ \left.\dfrac{\partial f_2}{\partial P}\right|_{(P_0,G_0)} (P - P_0) + \left.\dfrac{\partial f_2}{\partial G}\right|_{(P_0,G_0)} (G - G_0) \end{bmatrix}.$$

This equation can be written in matrix-vector form:

$$\begin{bmatrix} \dfrac{dP}{dt} \\ \dfrac{dG}{dt} \end{bmatrix} = \begin{bmatrix} \left.\dfrac{\partial f_1}{\partial P}\right|_{(P_0,G_0)} & \left.\dfrac{\partial f_1}{\partial G}\right|_{(P_0,G_0)} \\ \left.\dfrac{\partial f_2}{\partial P}\right|_{(P_0,G_0)} & \left.\dfrac{\partial f_2}{\partial G}\right|_{(P_0,G_0)} \end{bmatrix} \begin{bmatrix} P - P_0 \\ G - G_0 \end{bmatrix}.$$

Now we define vectors $\vec{X}(t)$ and $\vec{X}_0$ and matrix $J(\vec{X}_0)$ as

$$\vec{X}(t) = \begin{bmatrix} P(t) \\ G(t) \end{bmatrix}, \quad \vec{X}_0 = \begin{bmatrix} P_0 \\ G_0 \end{bmatrix} \quad \textbf{and} \quad J(\vec{X}_0) = \begin{bmatrix} \left.\dfrac{\partial f_1}{\partial P}\right|_{(P_0,G_0)} & \left.\dfrac{\partial f_1}{\partial G}\right|_{(P_0,G_0)} \\ \left.\dfrac{\partial f_2}{\partial P}\right|_{(P_0,G_0)} & \left.\dfrac{\partial f_2}{\partial G}\right|_{(P_0,G_0)} \end{bmatrix}.$$

Using these, the linearised system becomes

$$\frac{d\vec{X}}{dt} = J(\vec{X}_0)\left(\vec{X}(t) - \vec{X}_0\right).$$

Matrix $J(\vec{X}_0)$ is a matrix with elements that do not depend on time t. It is called the *Jacobian matrix* and plays an important role in our analysis of the system.

Exercise B

1/1 point (graded)

Keyboard Help

PROBLEM

Construct the Jacobian matrix at the equilibrium (100,0) for our system of differential equations.

$$J\left(\begin{bmatrix}100\\0\end{bmatrix}\right) = \begin{bmatrix}-0.7 & -4\\0 & 0.55\end{bmatrix}$$

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FEEDBACK

✔ Correctly placed 4 items.

✔ Good work! You have correctly constructed the Jacobian matrix.

✔ Your highest score is 1.0

The linearised system is now in matrix-vector form, and we do one final step and that is to give the combination $\vec{X} - \vec{X}_0$ another shorter name, $\vec{M}$:

$$\vec{M}(t) = \vec{X}(t) - \vec{X}_0 = \begin{bmatrix}P(t)\\G(t)\end{bmatrix} - \begin{bmatrix}P_0\\G_0\end{bmatrix}.$$

The time derivative of this vector is easily calculated:

$$\frac{d\vec{M}}{dt} = \frac{d\vec{X}}{dt} - \frac{d\vec{X}_0}{dt} = J(\vec{X}_0)(\vec{X}(t) - \vec{X}_0) - \vec{0} = J(\vec{X}_0)\,\vec{M}(t).$$

So, the time derivative of $\vec{M}$ is the Jacobian matrix at the equilibrium point multiplied by $\vec{M}$ itself:

$$\frac{d\vec{M}}{dt} = J(\vec{X}_0)\,\vec{M}(t).$$

On the next page, you will learn what this very compact system of linearized differential equations can tell you about the original differential equations.

English ▼

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